

MATH102BKK · HARBOUR.SPACE BANGKOK

Intro to Higher Math

From getting the answer to knowing why the answer is right.

Session	Date	Today
1 of 15	Mon 7 Sep 2026	Overview + refresher

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Software Engineers at Autopilot — and we have done all of this together.

2017 – 2023

School №199, Tbilisi

Physics & Mathematics

since 2022

Teaching mathematics

Privately, then here

2023 – 2026

Harbour.Space @ UTCC

CS & Data Science, double major

since 2023

ICPC

Competing for Harbour.Space

since 2025

Brainstormers

Co-founded — tournaments in logic

2026

Autopilot

Fintech and AI systems

Who you are

1

What is your name?

And what you would like to be called in class.

2

Where are you from?

Country, city — and how long you have been in Bangkok.

3

What is your background in maths?

Last course you took, and how you feel about proofs.

WHY WE ASK

This class runs on discussion and group work. Knowing the room — and hearing that other people are also new to proof — is what makes people willing to speak.

School maths ends where university maths begins

What you were rewarded for

- Getting the number at the bottom of the page
- Recognising which technique to apply
- Speed and accuracy

What you will be rewarded for here

- A claim, stated precisely
- An argument that forces the claim to be true
- Writing it so another person is convinced

THE SHIFT A calculation can be checked. An argument must be **understood**. Everything in the next three weeks trains the second skill.

The course at a glance

3

weeks · 7–25 Sep 2026

45

contact hours · 3 h/day

4

ECTS credits

Where it takes you next

Calculus · Linear algebra · Discrete mathematics · Graph theory · Probability — every one of them assumes the language you build here.

Prerequisites

None formally. A solid high-school background is assumed, and today's second half checks it is actually there.

Books

Simmons, *Precalculus Mathematics in a Nutshell* · Barrington, *A Mathematical Foundation for Computer Science* · Gibilisco, *Algebra Know-It-All*

Fifteen sessions, one argument

Week 1 · Language

1. Overview + refresher
2. Sets & set operations
3. Functions
4. Function families
5. Proof techniques

Week 2 · Method

6. Mathematical induction
7. Combinatorics + binomial theorem
8. **Midterm**
9. GCD & divisibility
10. Modular arithmetic

Week 3 · Structure

11. Vectors in $\mathbb{R}^n$
12. Polynomials
13. Complex numbers
14. Recap · maths in the AI era
15. **Final exam**

How the grade is built

COMPONENT	WEIGHT
Homework assignments	60%
Final exam	20%
Midterm exam	10%
Participation & in-class activities	10%

Homework dominates deliberately: proof is a skill, and skills are built by repetition, not by one exam morning.

MARKING RULE

A correct final answer with no reasoning is worth **very little**.

A wrong final answer with a clear, nearly-correct argument is worth **a lot**.

Collaboration, AI, and the one hard line

ENCOURAGED

Discuss the material. Compare strategies. Argue about solutions. Use AI tools as a learning aid.

PROHIBITED

Copying solutions from other students or external sources and submitting them as your own.

THE TEST THAT SETTLES IT

You must be able to **explain and justify every solution you submit**, on the spot, without notes. If you cannot, it is not yours yet — whoever or whatever produced it.

PART II

School math refresher

Not revision. We take four things you already compute and ask what makes them true — which is the habit the rest of the course is built on.

2

The language we will use

SYMBOL	READS AS
$\in$	is an element of
$\subseteq$	is a subset of
$\implies$	implies
$\iff$	if and only if
$\forall$	for all
$\exists$	there exists
$\mathbb{N} \mathbb{Z} \mathbb{Q} \mathbb{R}$	naturals, integers, rationals, reals
$\sum_{k=1}^n$	the sum over $k = 1$ to n

DEFINITION $P \implies Q$ asserts that there is no case in which P holds and Q fails. When P is false the implication is true – *vacuously*.

TRAP $P \implies Q$ is not $Q \implies P$. " $x = 2 \implies x^2 = 4$ " is true; its converse fails at $x = -2$.

Every rule is a theorem with a hypothesis

LAWS OF EXPONENTS For $a > 0$ and all real m, n :

$$a^m a^n = a^{m+n} \quad (a^m)^n = a^{mn} \quad a^{-n} = \frac{1}{a^n}$$

WHY $a^0 = 1$ Put $n = 0$ in the first law: $a^m a^0 = a^m$. As $a \neq 0$ we may divide by a^m , giving $a^0 = 1$. It is **forced** by the law, not chosen for convenience. ■

TRAP $\sqrt{x^2} = x$ is false. The radical returns the non-negative root, so $\sqrt{x^2} = |x|$. Test $x = -3$.

WORKED
$$\frac{8^{2/3} \cdot 2^{-1}}{4^{1/2}} = \frac{4 \cdot \frac{1}{2}}{2} = 1$$

Spot the missing hypothesis

ACTIVITY 1 GROUPS OF THREE

12 MIN

Every statement below is false as written. For each one, give a counterexample, then repair it by adding the **smallest** hypothesis that makes it true – or argue that it cannot be repaired.

1. $(a^m)^n = a^{mn}$

2. $\sqrt{x^2} = x$

3. $\sqrt{ab} = \sqrt{a} \sqrt{b}$

4. $\log(a + b) = \log a + \log b$

5. $\frac{1}{a+b} = \frac{1}{a} + \frac{1}{b}$

Share: each group presents one repair and the counterexample that forced it.

Absolute value is distance

DEFINITION

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Read $|a - b|$ as the distance from a to b on the line.

PROPOSITION

For all real x, y and every $c > 0$:

$$|xy| = |x||y| \quad |x| \leq c \iff -c \leq x \leq c$$

$$|x + y| \leq |x| + |y| \quad (\text{triangle inequality})$$

WORKED

$|2x - 3| < 5$ says $2x$ is within 5 of 3, so $-2 < 2x < 8$, that is $-1 < x < 4$. Every step is an **equivalence**.

Distance on the line

ACTIVITY 2 PAIRS

10 MIN

Turn each sentence into an absolute-value inequality, solve it, and draw the solution set on a number line.

1. x is within 3 of 7.
2. x is at least 2 away from -1 .
3. $2x$ is closer to 5 than 1 is.

Share: two pairs draw their number lines on the board; the room decides which endpoints are included, and says why.

Order is three axioms; the rest is theorems

ORDER AXIOMS For all real a, b, c :

1. exactly one of $a < b$, $a = b$, $b < a$ holds;
2. $a < b \implies a + c < b + c$;
3. $a < b$ and $0 < c \implies ac < bc$.

EVERYTHING ELSE IS DERIVED Claim: $a < b \implies -b < -a$. By (2)

add $-a - b$ to both sides of $a < b$:

$$a - a - b < b - a - b \implies -b < -a$$

No new rule was needed. This is what "axiomatic approach" means. ■

What you may and may not do

OPERATION ON $A < B$	LEGAL?	TRACEABLE TO
Add c to both sides	Always	axiom (2)
Multiply by $c > 0$	Always	axiom (3)
Multiply by $c < 0$	Only if you flip	(3) applied to $-c$
Multiply by an unknown x	No	sign of x unknown – split cases
Square both sides	Only if both are ≥ 0	$-3 < 2$ but $9 > 4$

ROW 3, TRACED

$-2 < 3$. Multiply by -4 : apply axiom (3) with $c = 4$ to get $-8 < 12$, then negate using $a < b \Rightarrow -b < -a$: $8 > -12$.

CLASSIC

From $\frac{1}{x} < 1$ you may **not** conclude $x > 1$. Check $x = -2$: the premise holds, the conclusion fails.

Always, sometimes, or never

ACTIVITY 3 GROUPS OF THREE

12 MIN

You are given $a < b$. Sort these moves into **always legal**, **legal under a condition**, or **never legal**. State the condition, or give a counterexample.

1. Add c to both sides.
2. Multiply both sides by c .
3. Square both sides.
4. Take reciprocals of both sides.
5. Multiply both sides by x , where x is the unknown.
6. Apply a strictly increasing function f to both sides.

Share: each group defends one card. Where two groups disagree, the counterexample settles it at the board.

Where the quadratic formula comes from

DERIVATION Start from $ax^2 + bx + c = 0$ with $a \neq 0$ and divide by a :

$$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Complete the square on the left, using $x^2 + \beta x = \left(x + \frac{\beta}{2}\right)^2 - \frac{\beta^2}{4}$:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Take square roots – *both* of them – and subtract $\frac{b}{2a}$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

■

WHAT THE DISCRIMINANT DECIDES $b^2 - 4ac > 0$: two real roots. $= 0$: one repeated root. < 0 : none in $\mathbb{R}$ – which is exactly the gap Session 13 fills with $\mathbb{C}$.

Derive it without us

ACTIVITY 4 PAIRS

12 MIN

1. Solve $3x^2 - 12x + 7 = 0$ by completing the square. Do not use the formula.
2. Now close your notes and redo the derivation for a general $ax^2 + bx + c = 0$.
3. Write one sentence saying what $b^2 - 4ac$ tells you, and why it tells you that.

Share: one pair derives the general case at the board while the room checks that every step is an equivalence.

On your own, before you leave

1 Solve $|3x + 1| \geq 4$ and name which steps are equivalences.

2 Find all real x with $\frac{2}{x-1} \leq 1$, handling the sign of $x - 1$ explicitly.

3 Show $x^2 + y^2 \geq 2xy$ for all real x, y , and say exactly when equality holds.

RULE FOR TODAY One sentence of justification under every answer. Hand it in on the way out — it is not graded, it tells us who to sit with tomorrow.

What today was actually about

- The rules you memorised are **theorems with hypotheses**. Learn the hypotheses.
- Order has three axioms; every inequality move must trace back to them.
- Marks live in the justification, not the answer.

hypothesis

counterexample

equivalence step

axiom

vacuous truth

Homework 1

12 diagnostic items — algebra, absolute value, inequalities. Each answer carries one sentence of justification.

Next session: **Sets & set operations** — the notation the whole course is written in.